

The Deixis Manual

Notes, Highlights, And Connections

Mark a passage. Share its explanation. Draw the relationship.

DEVELOPMENT 0.2.0 TYPST 0.15.0+

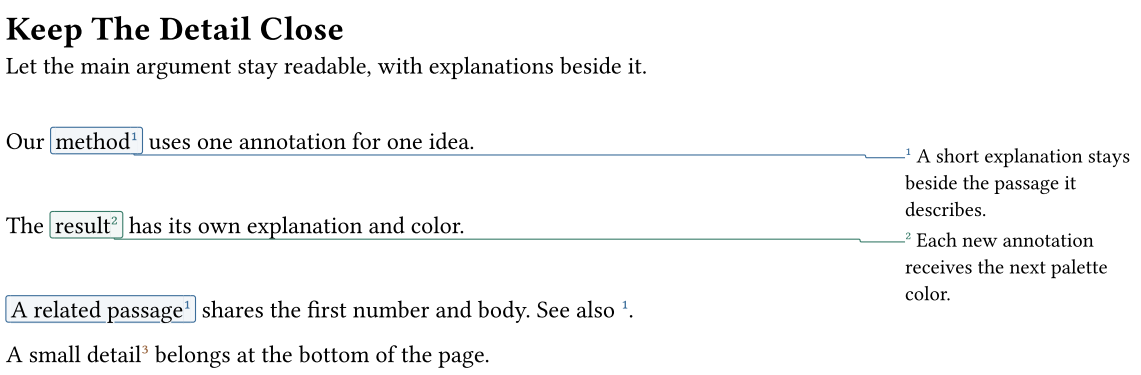
A visual guide with worked examples and a compact API reference.
Detailed recipes and complete record contracts live in `docs/guides`.

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Annotating With Deixis

Deixis brings notes, spatial highlights, and visual connections to Typst. It began with presentation annotations and now also serves technical documents, editorial notes, and drawings made from ordinary document content.
Start with one marked passage and one explanation. Give the annotation a name when other passages should share it. Add a drawn connection when the relationship needs to be visible on the page.



First Notes: two colored margin annotations, one reused passage, and a native footnote. The editable source is `examples/first-notes.typ`.

The package runtime needs no external drawing library. It includes native routes and arrowheads, while containers, body renderers, and connector painters remain customizable. Public document APIs share one identity and scope model.

This manual describes the **unreleased 0.2.0 checkout**, requiring **Typst 0.15.0+**. The downloadable `docs/manual.pdf` is built and checked from this source. For rebuild and release commands, see `docs/development/release.md`. A compact reference to all public APIs follows this tour; the task-oriented guides in `docs/guides` cover advanced integration and limits.

First Steps

Install The Checkout

From the repository, install the development package locally:

```
python3 scripts/install.py --namespace preview --link --reinstall
```

This makes `@preview/deixis:0.2.0` available to Typst CLI and VS Code/Tinymist. That version has not been published to Typst Universe. The local override keeps the old 0.1.1 cache separate; see `CONTRIBUTING.md` for alternatives.

Write A Note

```
#import "@preview/deixis:0.2.0": *
#show: deixis-setup-notes

A useful result#deixis-footnote[A short explanation.]
is easier to read when its details have a place.
```

The wrapper creates a numbered mark and its body; no public ID is needed here. Footnotes use the native page footer: long bodies paginate and look like `std.footnote` entries, even when declared in a heading. Source indentation in a body folds away, while explicit paragraph breaks remain. A long bordered body needs a breakable `block` container rather than `rect`. See `docs/guides/footnotes.md` and `examples/long-footnote.typ`.

Move An Explanation Into The Margin

Set the page geometry before setup, leaving room for the body:

```
#import "@preview/deixis:0.2.0": *
#set page(margin: (left: 25mm, right: 50mm, y: 25mm))
#show: deixis-setup-notes

Our #deixis-margin-note(id: <method>, width: 38mm, side: right,
  connector: "right-angle",
)[method][The explanation stays beside the passage it describes.]
keeps the main text short.
```

Two content arguments supply the marked passage and explanation. With one content argument, a wrapper creates a solo numbered mark and uses that argument as the explanation. Omit `connector:` for a note without a drawn connector.

Marks, Bodies, And Connections

An annotation has one original mark, one number, and at most one attached body. The original owns its logical identity and counter; the body owns its explanation, formatting, and placement policy. A connection owns how two endpoints are joined.

Share One Explanation

Use `id:` to create the original; use `of:` to attach a body or reuse the mark:

```
#deixis-mark(id: <detail>)[First passage]
#deixis-margin-note-body(of: <detail>)[One explanation.]
#deixis-mark(of: <detail>)[A related passage]
See also @detail.
```

The original, body, and reused marks may appear in any order. Only the original advances the counter. Another `id:` with the same name is an error. A body-only `id:` creates a standalone explanation without a numbered original. A reused mark adds a return-link origin and keeps the original number, body, and captured mark style. It does not add another drawn connector. Set `backlink: true` on the note/body to show return links. See `docs/guides/identities.md` and `docs/guides/shared-notes.md`.

Add A Connection Deliberately

A margin or inset body draws one connector from its original mark: a route name is enough, as in `connector: "curve"`, and a record adds detail, as in `connector: (route: "curve", heads: (body: "stealth"))`. For a connection from another marked passage, name that occurrence:

```
#deixis-mark(of: <detail>, occurrence: <again>)[Another passage]
#deixis-connect(
  from: (mark: <detail>, occurrence: <again>), to: <detail>,
  connector: (route: "curve", heads: (body: "stealth")),
)
```

Independent connections can target accepted margin/inset body frames, framed inline bodies, live body points, pins, or attached objects. They neither create nor move a body. Connections use their own foreground overlay by default; `connector.layer` can select background independently of the body's layer.

Choose A Placement Policy

Identity lookup and placement are different decisions. Declaring a body before its original does not make every adapter follow that mark to another page.

Body	What Selects Its Position?
Margin, overlay	The original mark or an explicit anchor; packing may spill to later existing pages.
Margin, flow	Immediate placement on the declaration page, with no collision packing or cross-page relocation.
Page footnote	The native footer, allocated at the body declaration. Long bodies paginate.
Inset	The declaration page by default, with optional anchors, offsets, or a placement callback.
Inline	Native flow: right after its passage for the wrapper, or where a `-body` call appears.
Endnote	An eligible `deixis-print-endnotes` call.

Immediate/native flow is useful when the body belongs in normal layout or must stay on its declaration page. Native placement can affect the surrounding paragraph; immediate margin notes do not use overlay packing.
Foreground/background overlays resolve document geometry before painting. Overlay-only declarations add no text-flow space. Overlays can use existing pages, but cannot create new pages. Within an overlay, `z-index` chooses drawing priority; the body and its connectors have independent priorities.
Collected flow lets native layout own a local collection and its pagination: block footnotes collect at the block's end, while endnotes use explicit printers. See `docs/guides/ownership.md`, `rendering.md`, and `blocks.md`.

Configure Appearance

```
#deixis-set(palette: (rgb("3B6D9A"), rgb("387C69"), rgb("9A663F")))
```

Each annotation captures one accent; reuse does not consume another color. Explicit styles win. `deixis-set` applies to subsequent declarations; a `deixis-block` provides a local settings scope, while ordinary Typst brackets do not. Body marker styles resolve at the body declaration; reused marks keep the original mark style. See `docs/guides/configuration.md` and `appearance.md`.

Drawings, Slides, And Next Steps

Route Your Own Objects

The same routing core works without notes or numbering. Register content where it is drawn, then declare each connection independently:

```
#import "@preview/deixis:0.2.0": *
#show: deixis-setup-notes
#show: deixis-routing.with(policy: "avoid")

#grid(columns: (auto, 1fr, auto),
  deixis-attach(id: <source>)[Source], [],
  deixis-attach(id: <target>)[Target],
)
#deixis-connect(from: (node: <source>), to: (node: <target>),
  connector: (heads: (body: "stealth")))
```

`manual` keeps authored routes; `avoid` searches around registered obstacles; `separate` also considers nearby connectors. Objects stay fixed. Pin ports, exact waypoints, and optional cuts/crossing gaps refine the drawing without replacing the connector model. For known coordinates, `deixis-route` returns native geometry without document setup. Start with `examples/routing-in-place.typ` and `docs/guides/document-routing.md`.

Integrate With A Theme

Configure native foreground/background before applying setup. After replacing either later, reapply `deixis-setup-notes`; existing identities and counters persist. Host-owned themes can install explicit overlay hooks. Changing page dimensions or margins alone does not require setup again. See `docs/guides/setup.md`.
Touying reveals retain hidden annotation ownership, but repeated subslides need frame-local identities for explicit IDs/references. Use the maintained examples and `docs/guides/visibility.md` / `touying-identities.md` for that integration.

Read By Task

- **Prepare a final copy:** `review-comments.md`; write `deixis-comment` notes and use `deixis-set(omit: (series: ("review",)))` to keep passages without them. Omitted originals consume no numbers or colors; attached bodies/reused marks follow them. References to omitted IDs report an error. `omit: none` includes subsequent series; a Deixis block restores its parent's policy.
- **Style annotations:** `appearance.md`, `configuration.md`, `containers.md`.
- **Write long documents:** `collected-notes.md`, `blocks.md`, `counters.md`.
- **Customize drawing:** `connectors.md`, `document-routing.md`, `numeric-routing.md`.
- **Build an index:** `queries.md`, `outlines.md`, `note-data.md`.
- **Edit a text or build a glossary:** `editorial-recipes.md`, `review-comments.md`.
- **Replace body output:** `custom-renderers.md`, with live destination/content helpers.
- **Migrate or check limits:** `compatibility.md`.

These guides live in `docs/guides`; `docs/README.md` is the task-oriented index. The compact reference follows this tour. A searchable documentation website can be built locally from the guides and source API catalog; hosting remains undecided.

Worked Examples

Read the source on the left and compare the result on the right. Each example is compiled as its own document, with independent labels, counters and page geometry. The SVG previews show actual output; they do not execute notes inside this manual. Open the named source for its full setup and working PDF links.
The specimens import Deixis and use a muted accent. “No Setup” drawings omit the final show rule:

```
#import "@preview/deixis:0.2.0": *
#let teal = rgb("32736D")
#show: deixis-setup-notes
```

Page size/margins and text size are chosen for the small specimen. Margin-note examples reserve a wide right margin. The displayed source is the exact authored body; page geometry and the common preamble are in the named editable file. For Touying, its theme/overlay configuration is in the complete slide example.
The Touying comparison gives both frames the same scale in a wider output column.
The following pages progress from ordinary notes to shared identities and styling, then custom content and drawings. Optional effects are shown explicitly; they are not enabled by ordinary note calls.

A Native Footnote

SETUP REQUIRED EDITABLE SOURCE · examples/footnote.typ

SOURCE

```
A useful result#deixis-footnote[A short
explanation.]
is easier to read.
```

RENDERED OUTPUT

A useful result[1] is easier to read.

[1] A short explanation.

A single content argument creates a solo mark and its explanation. Native flow reserves the footer slot.

Choose An Inline Mark

SETUP REQUIRED EDITABLE SOURCE · examples/inline-modes.typ

SOURCE

```
#for mode in ("box", "highlight",
  "underline", "parentheses",
  "text-fill", "none") {
  deixis-mark(inline-mode: mode,
    stroke: teal)[A marked passage]
  linebreak()
}
```

RENDERED OUTPUT

A marked passage
A marked passage
A marked passage
A marked passage
A marked passage
A marked passage

Decoration changes the passage without adding a note body. These anonymous marks have no number.

One Body, Several Passages

SETUP REQUIRED EDITABLE SOURCE · examples/shared-endnote.typ

SOURCE

```
An idea#deixis-endnote(id: <idea>)[
  One explanation for both uses.
] appears again#deixis-mark(of: <idea>).

A new idea#deixis-endnote[
  A second explanation.
] gets a new number.

#deixis-print-endnotes()
```

RENDERED OUTPUT

An idea[1] appears again[1].

A new idea[2] gets a new number.

1 One explanation for both uses.

2 A second explanation.

Reuse keeps the original number and body; the next original advances the counter. The printer chooses the body position.

Declare The Body First

SETUP REQUIRED EDITABLE SOURCE · examples/forward-margin.typ

SOURCE

```
#deixis-margin-note-body(
  of: <later>, width: 70pt,
  side: right,
)[A forward explanation.]

Our #deixis-mark(id: <later>)[method]
has an explanation.
```

RENDERED OUTPUT

Our method[1] has an explanation.

[1] A forward explanation.

Forward lookup resolves identity. An overlay margin follows the original even when its body was declared earlier.

Cycle Annotation Colors

SETUP REQUIRED EDITABLE SOURCE · examples/palette.typ

SOURCE

```
#deixis-set(palette: (
  rgb("3B6D9A"), rgb("387C69"),
  rgb("9A663F"),
))
#for i in range(3) {
  deixis-margin-note(width: 70pt,
    side: right)[Passage][
    Explanation #(i + 1).
  ]
  v(20pt)
}
```

RENDERED OUTPUT



Each original takes one accent. Shared marks retain it and explicit component styles can override it.

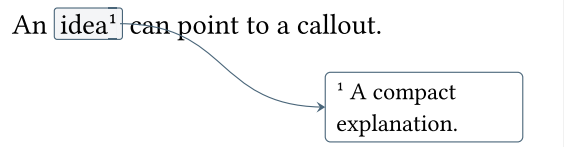
A Callout With An Arrow

SETUP REQUIRED EDITABLE SOURCE · examples/inset.typ

SOURCE

```
An #deixis-inset-note(
  layer: "foreground", width: 90pt,
  dx: 95pt, dy: 38pt,
  connector: (route: "curve",
    heads: (body: "stealth")),
)[idea][A compact explanation.]
can point to a callout.
```

RENDERED OUTPUT



The mark and body use one wrapper. Placement offsets belong to the body; the connector has its own route and head.

Highlight A Formula

SETUP REQUIRED EDITABLE SOURCE · examples/region.typ

SOURCE

```
#deixis-endnote(
  mark-type: "region", inline: true,
  layer: (mark: "background"),
  fill: teal.lighten(92%),
  stroke: teal,
)[$a^2 + b^2 = c^2$][
  A relation worth explaining.
]

#deixis-print-endnotes()
```

RENDERED OUTPUT

$a^2 + b^2 = c^2$[1]

1 A relation worth explaining.

A region wrapper measures the actual content. Mark and body layers can be selected independently.

Attach Region Corners

SETUP REQUIRED EDITABLE SOURCE · examples/pins.typ

SOURCE

```
#deixis-attach(prefix: none,
  pins: (
    begin: (anchor: top + left),
    end: (anchor: bottom + right),
  ),
  rect(inset: 12pt, stroke: none)[
    $sum_(i=1)^n i = (n(n+1))/2$
  ],
)
#deixis-region-mark(
  pins: (<begin>, <end>),
  padding: 3pt, stroke: teal,
)
```

RENDERED OUTPUT

$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$

Pins can be attached around an existing object. Their padding is geometric: it adds no space to the object.

Restore Local Settings

SETUP REQUIRED EDITABLE SOURCE · examples/scoped-settings.typ

SOURCE

```
#deixis-set(stroke: blue)
#deixis-mark[Outer Style]

#deixis-block(inset: 5pt)[
  #deixis-set(stroke: orange)
  #deixis-mark[Local Style]
]

#deixis-mark[Outer Style Again]
```

RENDERED OUTPUT

Outer Style

Local Style

Outer Style Again

A Deixis block scopes its settings and numbering. After it ends, the outer defaults resume.

Keep Notes With A Table

SETUP REQUIRED EDITABLE SOURCE · examples/local-notes.typ

SOURCE

```
#deixis-block[
  #figure(caption: [Measurements.])[
    #table(columns: (1fr, 1fr),
      [Sample], [Result],
      [Control], [12.4],
      [Trial], [18.7#deixis-footnote(
        numbering: "a",
      )[Excludes one damaged sample.]],
    )
  ]
]
```

RENDERED OUTPUT

Sample	Result
Control	12.4
Trial	18.7[a]

Table 1: Measurements.

[a] Excludes one damaged sample.

One block owns the whole figure and its caption. Its footnotes collect below the object with independent numbering.

Give A Body A Container

SETUP REQUIRED EDITABLE SOURCE · examples/container.typ

SOURCE

```
A detail#deixis-endnote(
  container-func: deixis-rect-container,
  container-args: (inset: 7pt),
  stroke: (mark: none, body: teal),
)[A boxed explanation.]

#deixis-print-endnotes(
  render-group:
    deixis-native-render-group,
)
```

RENDERED OUTPUT

A detail[1]

1 A boxed explanation.

Container options stay with the body. A stacked endnote group delegates to its single-note container.

Customize The Body Layout

SETUP REQUIRED EDITABLE SOURCE · examples/custom-renderer.typ

SOURCE

```
#let card(n, ..args) = block(
  inset: 8pt, fill: teal.lighten(94%),
  stroke: (left: teal + 2pt),
)[
  #deixis-generate-body-marker(n)
  #deixis-generate-body-content(n)
]

#deixis-endnote(render-single: card)[
  A custom card with native links.
]
#deixis-print-endnotes(
  render-group:
    deixis-native-render-group,
)
```

RENDERED OUTPUT

1

1 A custom card with native links.

The marker helper supplies the destination; the content helper retains live spans. Use each once.

Build A Custom Note List

SETUP REQUIRED EDITABLE SOURCE · examples/catalogue.typ

SOURCE

```
#deixis-endnote(id: <method>,
  data: (title: [Explain The Method]),
)[The method needs context.]
#deixis-mark(of: <method>)
#deixis-print-endnotes()

#deixis-note-outline(
  title: [Review Topics],
  summary: n => n.data.title,
)
```

RENDERED OUTPUT

[1]

1 The method needs context.

Review Topics

[1] Explain The Method 1

Author data stays separate from identity and layout. Query records can supply custom catalogues without replaying body content.

Connect Ordinary Content

SETUP REQUIRED EDITABLE SOURCE · examples/attached-nodes.typ

SOURCE

```
#grid(columns: (1fr, 1fr),
  gutter: 45pt,
  deixis-attach(id: <a>,
    rect(inset: 5pt)[Source]),
  deixis-attach(id: <b>,
    rect(inset: 5pt)[Target]),
)
#deixis-connect(
  from: (node: <a>),
  to: (node: <b>),
  connector: (heads: "stealth"),
)
```

RENDERED OUTPUT



Node IDs register bounds where objects are rendered. No detached list of object positions is needed.

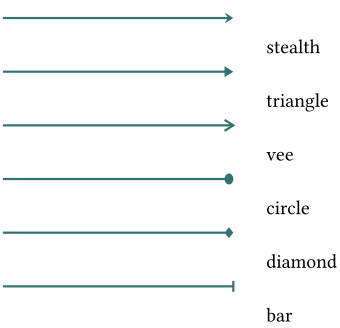
Choose A Native Arrowhead

NO SETUP EDITABLE SOURCE · examples/head-shapes.typ

SOURCE

```
#for shape in ("stealth", "triangle",
  "vee", "circle", "diamond", "bar") {
  context {
    let path = deixis-route(
      from: (0pt, 0pt),
      to: (105pt, 0pt))
    let paint = deixis-prepare-connector(
      path, stroke: teal + 1pt,
      heads: (body: shape))
    box(width: 120pt, height: 16pt,
      deixis-render-connector(paint))
    text(size: 9pt, shape)
    linebreak()
  }
}
```

RENDERED OUTPUT



Head size, fill, stroke and trimming are configurable. Numeric geometry and its painter work without note setup.

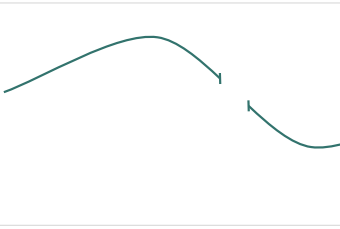
Cut A Curve Without Moving It

NO SETUP EDITABLE SOURCE · examples/curve-cuts.typ

SOURCE

```
#context {
  let path = deixis-route(
    from: (0pt, 25pt),
    to: (210pt, 25pt),
    route: "curve",
    via: ((70pt, 0pt), (140pt, 50pt)))
  let paint = deixis-prepare-connector(
    path, stroke: teal + 1pt,
    heads: (body: "stealth"),
    cuts: (at: 50%, length: 18pt),
    gap-marks: "slash")
  box(width: 220pt, height: 70pt,
    place(top + left, dy: 25pt,
      deixis-render-connector(paint)))
}
```

RENDERED OUTPUT



Cuts hide intervals on the completed curve. Optional indicators are independent; the remaining pieces retain the original geometry.

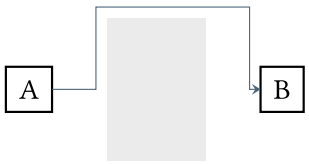
Route Around An Obstacle

SETUP REQUIRED EDITABLE SOURCE · examples/avoidance.typ

SOURCE

```
#show: deixis-routing.with(
  policy: "avoid", clearance: 5pt)
#let node(id, body) = deixis-attach(
  id: id, rect(inset: 6pt, body))
#grid(columns: (auto, auto, auto),
  gutter: 25pt, align: horizon,
  node(<a>)[A],
  deixis-attach(behavior: "avoid",
    rect(width: 45pt, height: 65pt,
      fill: luma(235), stroke: none)),
  node(<b>)[B],
)
#deixis-connect(from: (node: <a>),
  to: (node: <b>),
  connector: (heads: "stealth"))
```

RENDERED OUTPUT



The wrapper captures a policy for individually declared connections. Registered nodes stay fixed; the sharp orthogonal route avoids the middle rectangle.

Continue On Another Page

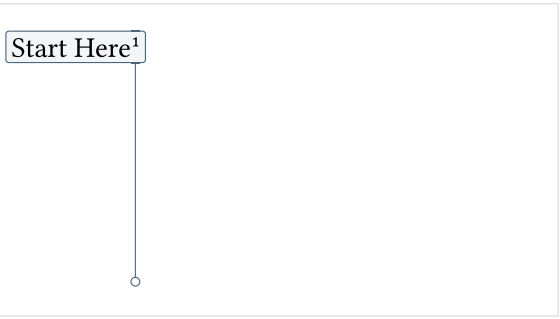
SETUP REQUIRED EDITABLE SOURCE · examples/page-portals.typ

SOURCE

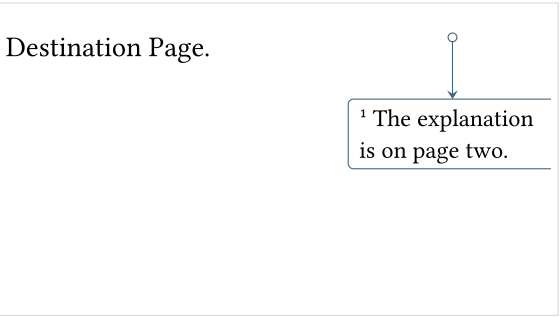
```
#deixis-mark(id: <across>)[Start Here]
#pagebreak()
#deixis-inset-note-body(of: <across>,
  layer: "foreground", width: 95pt,
  dx: 95pt, dy: 40pt,
  connector: (route: "orthogonal",
    continuation: "portal",
    heads: (body: "stealth")),
)[The explanation is on page two.]
Destination Page.
```

RENDERED OUTPUT

Page 1



Page 2



A portal marks each accepted page boundary. The two fragments use each page's geometry; no line is drawn across the intervening document.

Follow Endnotes In The Header

SETUP REQUIRED EDITABLE SOURCE · examples/adaptive-header.typ

SOURCE

```
#let header = context {
  let notes = deixis-query-notes(
    include-types: "endnote",
    page: here().page())
  if notes.len() > 0 {
    let pages = notes.map(n =>
      n.marks.map(m =>
m.dest.page())
    ).flatten().dedup().sorted()
    text(size: 9pt)[
      Notes For Pages
      #pages.map(str).join([, ])
    ]
  }
}
#set page(header: header)

Our #deixis-endnote(
  id: <method>,
)[
  method
][An explanation of the method.]
needs context.
#pagebreak()
Use the #deixis-mark(
  of: <method>,
)[
  method again
].
#deixis-print-endnotes()
```

RENDERED OUTPUT

Page 1

Our method[1] needs context.

1

Page 2

Notes For Pages 1, 2

Use the method again[1].

1 An explanation of the method.

2

The header reads complete body spans on its physical page, then gathers all original and reused mark pages. Querying never renders an explanation again. These source page numbers are physical; native page-label formatting is a separate step.

Reveal An Annotation With Its Passage

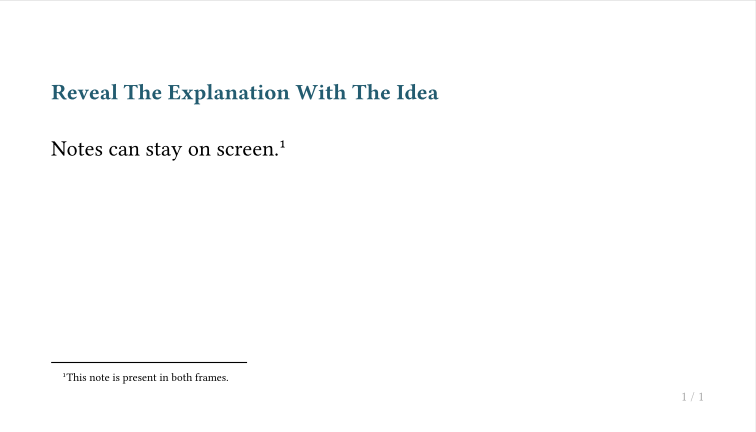
SETUP REQUIRED EDITABLE SOURCE · ../examples/touying-reveals.typ

SOURCE

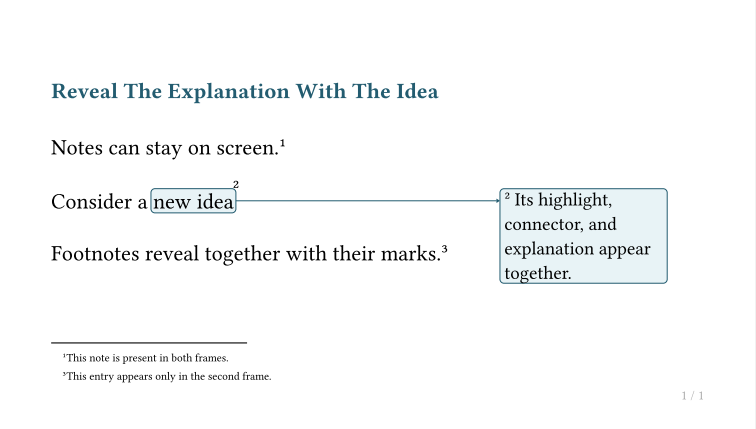
```
#pause
  #v(0.6em)
  Consider a #deixis-inset-note(
    mark-type: "region",
    inline: true,
    layer: (
      mark: "background",
      body: "foreground",
    ),
    width: 190pt,
    dx: 300pt, dy: 40pt,
    fill: rgb("e8f3f6"),
    stroke: rgb("235c70"),
    connector: (
      route: "curve",
      ports: (
        mark: "right",
        body: "left",
      ),
      heads: "triangle",
    ),
  )[new idea][
    Its highlight, connector,
    and explanation appear
    together.
  ]
```

RENDERED OUTPUT

Frame 1 · Before Pause



Frame 2 · After Pause



Touying renders two subslides. The second reveals the region, connector and callout together; hidden declarations retain their semantic ownership.

Separate Readings From Commentary

SETUP REQUIRED EDITABLE SOURCE · examples/editorial-apparatus.typ

SOURCE

```
#let apparatus(n, ..args) =
block[
  #deixis-generate-body-marker(n)
  #emph(n.data.lemma)#h(0.3em)
  #deixis-generate-body-
content(n)
]
#let source = deixis-
endnote.with(
  series: "sources", numbering:
"a",
  render-single: apparatus)
#let editor = deixis-
endnote.with(
  series: "editorial", numbering:
"1")

The #source(id: <reading>, data:
(lemma: [harbour],))[
  harbour][MS A; MS B reads
“haven”.]
was quiet. The #deixis-mark(of:
<reading>)[harbour]
returns in the next
stanza.#editor[The repetition is
deliberate.]

== Source Readings
#deixis-print-endnotes(series:
"sources",
  render-group: deixis-native-
render-group)
== Editorial Notes
#deixis-print-endnotes(series:
"editorial")
```

RENDERED OUTPUT

The harbour[a] was quiet. The harbour[a] returns in the next stanza.[1]

Source Readings

a *harbour* MS A; MS B reads “haven”.

Editorial Notes

1 The repetition is deliberate.

Independent series give source readings and editorial comments their own numbering. A custom renderer adds a lemma; reused passages share one reading.

Find Open Review Comments

SETUP REQUIRED EDITABLE SOURCE · examples/review-index.typ

SOURCE

```
#let comment = deixis-
endnote.with(series: "review")
The result is #comment(data: (
  author: "Ada", status: "open",
title: [Check Scope],
))[general][Does the argument
cover the boundary case?].
The #comment(data: (
  author: "Lin", status:
"resolved", title: [Name The
Source],
))[source][The citation has been
added.] is now identified.
#deixis-print-endnotes(series:
"review")

== Open Comments By Ada
#context {
  let pending = deixis-query-
notes(series: "review")
    .filter(n => n.data.author ==
"Ada"
      and n.data.status ==
"open")
  for n in pending {
    block(link(n.dest)[#n.marker
#n.data.title])
  }
}
```

RENDERED OUTPUT

The result is general[1]. The source[2] is now identified.

1 Does the argument cover the boundary case?

2 The citation has been added.

Open Comments By Ada

1 Check Scope

Author and status are ordinary annotation data. A filtered linked view selects open comments without removing resolved notes from the draft.

Link Terms To Their First Use

SETUP REQUIRED EDITABLE SOURCE · examples/glossary.typ

SOURCE

```
#deixis-mark(id: <lemma>, series:
"glossary",
  marker: none, data: (term:
"Lemma",))[*Lemma*]
means the dictionary form of a
word.
#deixis-endnote-body(of: <lemma>)
[A dictionary headword.]
Later, #context {
  let n = deixis-query-
notes(series: "glossary")
    .find(n => n.id == <lemma>)
  link(n.dest)[lemma]
} links to its first use.

#deixis-mark(id: <token>, series:
"glossary",
  marker: none, data: (term:
"Token",))[*Token*]
means one occurrence of a word.
#deixis-endnote-body(of: <token>)
[One occurrence in a text.]

== Glossary
#context {
  let terms = deixis-query-
notes(series: "glossary")
    .sorted(key: n =>
n.data.term)
  for n in terms {
    block[#strong(link(n.dest,
n.data.term)): #n.summary]
  }
}
```

RENDERED OUTPUT

Lemma means the dictionary form of a word. Later, lemma links to its first use.

Token means one occurrence of a word.

Glossary

Lemma: A dictionary headword.

Token: One occurrence in a text.

Explicit originals choose the first use. A sorted catalogue displays safe definition summaries without printing or executing their endnote bodies.

API At A Glance

Start with a note wrapper. Reach for marks/bodies when authoring positions differ, and independent connections when one explanation needs more edges. The entries below cover every public export, with shared/preset functions grouped together.

SETUP REQUIRED	Uses the document annotation system; apply `deixis-setup-notes` first.
NO SETUP	Works with supplied content or numeric geometry, without note setup.
CONDITIONAL	Numeric/supplied data needs no setup; document lookup or live annotation behavior does. See the entry.
INSTALLS SETUP	Installs the document machinery.
EXTENSION HOOK	Low-level callback or integration API; normal note calls usually suffice.

Badges classify **setup**, not Typst `context`. Queries and document geometry need context; relative styles and custom painting can need it even on a "No Setup" path. Direct render helpers do not initialize or register a note.
This is a compact reference: displayed parameters/defaults come from each entry's first function in runtime source. Long option sets are shared below; task guides explain records, edge cases and recipes. A listed `auto` is an argument default, not necessarily the final value: factory settings and the current Deixis scope can resolve it.

```
deixis-absolute-place
deixis-aligned-render-group
deixis-arrowhead
deixis-block
deixis-configure
deixis-ellipse-region
deixis-endnote-body
deixis-footnote-body
deixis-generate-body-content
deixis-generate-body-meta
deixis-inline-mark
deixis-inline-note-body
deixis-inset-note-body
deixis-margin-note-adaptive-layout
deixis-margin-note-exact-layout
deixis-mark
deixis-native-render-group
deixis-note-counter
deixis-page-overlay
deixis-pin
deixis-plain
deixis-prepare-connector
deixis-query-notes
deixis-rect-region
deixis-region-mark
deixis-render-head
deixis-route-group
deixis-routing-scene
deixis-setup-notes
deixis-sidenote-body
short
```

```
deixis-alert-container
deixis-aligned-render-single
deixis-attach
deixis-comment
deixis-connect
deixis-endnote
deixis-footnote
deixis-generate-backlinks
deixis-generate-body-marker
deixis-grid-render-group
deixis-inline-note
deixis-inset-note
deixis-margin-note
deixis-margin-note-body
deixis-margin-note-flow-layout
deixis-minipage
deixis-native-render-single
deixis-note-outline
deixis-phantom-mark
deixis-place-anchored
deixis-plain-container
deixis-print-endnotes
deixis-rect-container
deixis-ref
deixis-render-connector
deixis-route
deixis-routing
deixis-set
deixis-sidenote
deixis-update-note-counter
```

Shared Note Options

Wrappers take `[body]` or `[passage][body]`; body-only functions take `[body]`. Options apply only where accepted by that adapter. Inherited options stay omitted in the compact entries rather than repeated for every note type.

`id`	Original/standalone identity; omit for private wrapper identity.
`of`	Attach a body or reuse a mark; exclusive with id.
`data`	Opaque author fields on creation; inherited by attachments/reuse.
`target`	Rendering/counter scope. Attachments inherit their original instance; target: 0 deliberately selects the caller.
`series`	Counter/collection stream. Attachments inherit it; original wrappers choose their adapter's stream.
`marker`	Automatic numbering, a literal marker, or none to hide it.
`numbering`	Pattern/callback, or (restart: "page", pattern: "1") for original-page display ordinals. Assigned counters stay unchanged; reuse keeps the original marker.
`marker-style`	Callback for marker content; may select mark/body components.
`stroke / fill / radius`	Shared visual styles; component maps distinguish mark, body and connector.
`layer / z-index`	Placement/paint layer and within-layer priority; connector layer stays independent.
`backlink`	Return links from the body: true, false, always, never/none, or multiple.
`container-func / container-args`	Wrap the rendered body; native/custom options stay in container-args.
`render-single / render-group`	Replace body/group formatting; preserve live destination/content helpers.

Settings And Defaults

`deixis-configure` is an identical alias of `deixis-set`. For short calls, import the `short` module, then `import short: *`: use `configure`, `footnote` and `note-block`. Native `block` remains available. Other aliases omit `deixis-`; `short as dx` supports qualified calls such as `dx.footnote`.

`deixis-set` accepts palette, numbering, marker-style, body-style, backlink, marker-position, inline-mode, stroke, fill, radius, connector, separator, clearance, gap, marker-gap, indent, margin-layout, min-margin-width, side-strategy, mark-align, mark-align-strictness, spillover, overflow, region-shape, layer, z-index, render-single, render-group and container-func. Every argument defaults to `auto`, leaving current settings intact. Type/component dictionaries keep the API small; see `configuration.md`.

Factory marks use muted blue-gray and a faint tint. Body marker gaps are 0.05em for footnotes, 0.25em for margin/inset/inline, and 0.75em for endnotes. Inset and page-footnote bodies default to flow; margin bodies to foreground. Page footnotes use the native footer's typography, separator and gap. Read the placement table in the tour before changing layers. A palette is opt-in.

Connector Records

A margin/inset body accepts `connector: (...)`; `deixis-connect` uses the same record. A route name is shorthand, as in `connector: "curve"`. All fields are optional and merge with `deixis-set(connector: ...)` defaults field by field; enabling a head does not enable a disabled route.

`route`	Polyline, orthogonal/chamfer, interpolating curve/Catmull-Rom/Hobby, or B-spline influenced by controls. Route records use `kind` and algorithm-specific settings.
`ports`	Mark/body sides, side intervals, exact percentages, or a pin-defined custom socket. Outward normals describe departure/arrival direction.
`via / segments`	Exact visits or route-section records. Use one representation at a time; controls are distinct from exact waypoints.
`heads`	Independent mark/body heads: a native shape, head specification, callback, or none.
`stroke / radius`	Single line stroke and turn radius, independent of rectangular node edge/corner styles.
`cuts / gap-marks`	Hide intervals on finished curve arc length; optional slash/bar indicators decorate surviving gaps.
`crossings`	Optional automatic gaps on participating lower-priority shafts. It does not enable obstacle avoidance.
`continuation / spill`	Optional dot/portal/custom page markers and page-local allowed intervals. Margin destinations default to their physical channel, others to content width.
`scene`	Prepared scene or page callback for explicit interactions outside collected routing, or under manual policy.
`render`	Custom painter receiving prepared geometry, styles, heads and decorations. Delegate to deixis-render-connector.
`layer / z-index`	Foreground by default, or background; independent body/edge priorities. No connector flow layer.

Geometry And Failures

Pins and hard visits stay exact. Scene avoidance and collected repair require sharp orthogonal routes. Avoidance checks centerlines; thick strokes/heads may extend outside them. Node behaviors are avoid, ignore, cut or cross; indicators are always opt-in. Independent page fragments may have different continuation positions when page dimensions change. No shaft is drawn on intermediate pages.

The numeric API returns geometry/failure records; inspect failure status before preparing paint. Document APIs diagnose unresolved endpoints and unsupported geometry. Budgets bound repair work; they do not guarantee global optimality. See `connectors.md`, `document-routing.md`, and `routing-scenes.md` for details.

Setup And Defaults

Install setup before document annotations. Configuration can precede it. Reapply setup after replacing a native page foreground/background; identities and counters persist.

Guide: `docs/guides/setup.md`

deixis-setup-notes

INSTALLS SETUP Returns content → content

Document show rule: installs registration, references, visibility, validation and optional page overlays.

```
#show: deixis-setup-notes
```

SELECTED PARAMETERS · SOURCE DEFAULTS

```
override-page-footnotes = false
foreground = none
page-overlays = true
background = none
```

Set page geometry/layers first. Page footnotes default to native flow; override-page-footnotes: true makes the foreground overlay their default. Host themes may install both overlay hooks with page-overlays: false.

deixis-page-overlay

SETUP REQUIRED Returns content

Host integration hook for one page overlay; does not initialize the annotation registry.

```
deixis-page-overlay(layer: "foreground")
```

SELECTED PARAMETERS · SOURCE DEFAULTS

```
layer = "foreground"
```

Install background too: it records page geometry. See the setup guide for host-owned/scoped composition.

deixis-set
deixis-configure

NO SETUP Returns content

Updates subsequent defaults in the current Deixis scope; ordinary Typst brackets do not create a settings scope.

```
deixis-set(palette: "okabe-ito")
```

SELECTED PARAMETERS · SOURCE DEFAULTS

```
omit = auto
stroke = auto
connector = auto
render-single = auto
palette = auto
marker-style = auto
layer = auto
```

Top-level auto leaves settings unchanged. Type/component maps merge compatible branches; empty maps and branch auto have option-specific reset semantics. All accepted setting names are listed in the shared-options section. omit: (series: ("todo",)) removes selected annotations while preserving passages; none includes all series. `deixis-configure` is an identical alias; `short.configure` uses the same function.

Guide: `docs/guides/configuration.md`

short

NO SETUP Returns module

Opt-in module of short aliases. Use configure for settings and note-block for Deixis scopes; native block remains available.

```
#import short: *
```

Use short as dx for qualified calls. All function aliases preserve identity; configure replaces the keyword set and note-block preserves native block. See the getting-started guide for import syntax.

Guide: `docs/guides/getting-started.md`

Marks And Identity

Use id to create the original counter owner; of reuses it. Names accept nonempty strings or labels. An original, its one body and reused marks may appear in any order. Reuse does not add a drawn connector.

Guide: `docs/guides/identities.md`

deixis-mark

SETUP REQUIRED Returns content

Unified original/reused mark. Defaults to inline; pins select region mode. A call without id/of is anonymous decoration.

```
deixis-mark(id: <detail>)[A passage]
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`id` = none	`of` = none
`mark-type` = auto	`occurrence` = none
`marker` = auto	`numbering` = auto

mark-type: "region" highlights a passage or pins; "phantom" is a semantic origin. Set data on creation, not reuse. See shared options for styles, targets and region controls.

deixis-inline-mark

SETUP REQUIRED Returns content

Specialized inline form of deixis-mark, with an optional marked passage.

```
deixis-inline-mark(of: <detail>)[Another passage]
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`inline-mode` = auto	`marker-position` = auto

Box, highlight, underline, parentheses, text-fill and none are available. Reused marks retain the original styles beneath explicit overrides.

Guide: `docs/guides/shared-notes.md`

deixis-region-mark

SETUP REQUIRED Returns content

Highlights a passage or a pin-defined region, with page-local fragment geometry.

```
deixis-region-mark(inline: true)[$x^2 + y^2$]
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`pins` = ()	`padding` = auto
`layer` = auto	`inline` = false
`region-shape` = auto	

Supply passage or pins, never both. A flow region must fit one page. Foreground/background regions may span pages. Region marker numerals remain foreground.

Guide: `docs/guides/regions.md`

deixis-phantom-mark

SETUP REQUIRED Returns content

Creates/reuses an invisible semantic original; its body may still carry a numbered marker.

```
deixis-phantom-mark(id: <quiet>)
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`id` = none	`of` = none
`occurrence` = none	`marker` = auto

Phantom reuse accepts no passage or appearance overrides. An anonymous phantom draws nothing and consumes no palette slot.

Bodies And Collections

Wrappers accept [body] or [passage][body]. A -body call takes [body]. Wrapper id creates original + body; -body id creates a standalone body. With of, both forms attach only a body. Margin, inset and endnote bodies capture declaration typography; page footnotes use the native footer.

Guide: `docs/guides/ownership.md`

`deixis-footnote`
`deixis-footnote-body`

SETUP REQUIRED Returns content

Allocates a native footer slot at the body declaration, even when its original mark is on another page.

```
deixis-footnote[An explanation.]
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`layer` = `auto`	`backlink` = `auto`
`container-args` = `(:)`	

Native flow (the default) paginates long bodies and uses the footer's typography; separator/gap are native footnote.entry settings. Overlay footnotes reserve space but paint whole units. Local block footnotes collect at the block end. numbering: (restart: "page", pattern: "1") restarts displayed markers on each original page; assigned counts stay stable.

Guide: `docs/guides/rendering.md`

`deixis-margin-note`
`deixis-margin-note-body`

SETUP REQUIRED Returns content

Overlay bodies follow the original mark or anchor pin; native margin packing may spill onto later existing pages.

```
deixis-margin-note(width: 30mm, side: right)[Passage][Explanation.]
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`width` = `90%`	`side` = `auto`
`dx` = `auto`	`dy` = `0pt`
`spillover` = `auto`	`overflow` = `auto`
`mark-align` = `auto`	`mark-align-strictness` = `auto`
`anchor-pin` = `none`	

Leave a wide page margin. Flow places exactly on the declaration page with no collision packing. overflow: "error" checks the final frame. Connector shorthand selects the original mark.

Guide: `docs/guides/margin-overflow.md`

`deixis-sidenote`
`deixis-sidenote-body`

SETUP REQUIRED Returns content

Margin-note presets for the outside margin and loose mark alignment.

```
deixis-sidenote[A brief aside.]
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`side` = `"outside"`	`mark-align-strictness` = `"loose"`

Uses margin-note configuration/query type. Explicit arguments override the preset; mark-align-strictness: auto resumes configured strictness.

Guide: `docs/guides/configuration.md`

deixis-comment

SETUP REQUIRED Returns content

Review-comment preset: a margin note in the "review" series with lettered markers, author/status data and a bold header.

```
deixis-comment(author: "Ada", status: "open")[Passage][Comment.]
```

SELECTED PARAMETERS · SOURCE DEFAULTS

```
author = none                                   status = none
series = "review"                               numbering = "A"
header = auto
```

Omit the series for a final copy with deixis-set(omit: (series: ("review",))). Other options are margin-note options; of is rejected.
Guide: `docs/guides/review-comments.md`

deixis-inset-note
deixis-inset-note-body

SETUP REQUIRED Returns content

Callout whose page defaults to its body declaration; use anchors, offsets or a placement callback for deliberate positioning.

```
deixis-inset-note(dx: 100pt, dy: 40pt)[Passage][Callout.]
```

SELECTED PARAMETERS · SOURCE DEFAULTS

```
width = auto                                    layer = auto
placement = none                                dx = none
dy = none                                       anchor = (mark: horizon + right, body: horizon
                                                + left)
anchor-pin = none
```

Flow is the factory body layer. Overlay declarations are invisible in flow. The connector layer is independent. Custom containers receive container-args.

deixis-inline-note
deixis-inline-note-body

SETUP REQUIRED Returns content

An explanation in native flow: the wrapper places it right after its passage as an inline aside; a -body call places it where called.

```
deixis-inline-note[term][An inline explanation.]
```

SELECTED PARAMETERS · SOURCE DEFAULTS

```
inline = true                                   of = none
breakable = true                                container-args = (:)
```

The wrapper defaults to inline: true and keeps the sentence; inline: false gives a callout after the line (the -body default). Inline decoration bypasses block containers/custom single renderers. breakable: false supplies one frame for an explicit body endpoint.
Guide: `docs/guides/containers.md`

deixis-endnote
deixis-endnote-body

SETUP REQUIRED Returns content

Declares a collected explanation, rendered only by an eligible endnote printer.

```
deixis-endnote[A collected explanation.]
```

SELECTED PARAMETERS · SOURCE DEFAULTS

```
series = auto                                   backlink = auto
container-args = (:)
```

Wrapper series defaults to "endnote"; attachments inherit the original series. The printer controls where it appears. Body settings are retained until printing.
Guide: `docs/guides/collected-notes.md`

`deixis-print-endnotes`

SETUP REQUIRED Returns content

Prints unprinted eligible bodies in source order; successive printers do not duplicate output.

```
deixis-print-endnotes()
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`target` = `0`	`series` = `auto`
`after` = `none`	`before` = `none`
`render-group` = `auto`	`container-args` = `auto`

Omitted overrides retain body settings. Default grid bypasses single-note containers; select deixis-native-render-group to use them. after/before are exclusive source boundaries.

Guide: `docs/guides/collected-notes.md`

Connections And Drawing

A body connector is the short path for margin/inset notes. Independent edges select document endpoints without creating or moving them. Connections paint in their own foreground/background layer, below bodies at matching priorities.

Guide: `docs/guides/connectors.md`

deixis-connect

SETUP REQUIRED Returns content

Draws an independent edge between marks, bodies, attached objects or pins; declarations may precede endpoints.

```
deixis-connect(from: (node: <a>), to: (node: <b>))
```

SELECTED PARAMETERS · SOURCE DEFAULTS

```
from = none                                    to = none
connector = auto
```

Bare from IDs select original marks; bare to IDs select bodies. Use (mark: id, occurrence: name), (body: id, point: "end"), (node: id) or (pin: target) explicitly. Default route is straight, or sharp orthogonal inside deixis-routing.

Guide: `docs/guides/independent-connections.md`

deixis-routing

SETUP REQUIRED Returns content → content

Show-rule wrapper that captures a policy for individual connections and enabled margin/inset shorthand.

```
#show: deixis-routing.with(policy: "avoid")
```

SELECTED PARAMETERS · SOURCE DEFAULTS

```
policy = "separate"                            nodes = ()
clearance = 4pt                                spacing = 3pt
rounds = 2                                     max-searches = 64
```

manual keeps native routes; avoid searches independently; separate adds bounded group repair. Search needs sharp orthogonal geometry. Registered objects stay fixed. Nested wrappers form separate groups.

Guide: `docs/guides/document-routing.md`

deixis-route

NO SETUP Returns route / failure record

Builds source-local line/cubic geometry from finite absolute coordinates, optionally using a prepared scene.

```
deixis-route(from: (0pt, 0pt), to: (100pt, 40pt), route: "curve")
```

SELECTED PARAMETERS · SOURCE DEFAULTS

```
from = none                                    to = none
route = "straight-line"                        via = none
segments = none                                radius = 0pt
scene = none                                   page = none
```

Coordinates use x-right/y-down. origin is the source; segments are relative to it. No measurement or painting. Scene routes can fail: inspect status/detail before preparation. Avoidance requires sharp orthogonal paths.

Guide: `docs/guides/numeric-routing.md`

deixis-routing-scene

CONDITIONAL Returns scene record

Prepares reusable fixed rectangles/points in caller-defined page planes.

```
deixis-routing-scene(nodes: ((id: <a>, bounds: (left: 0pt, top: 0pt, right: 30pt, bottom:
20pt)),))
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`nodes` = `()`
`escape` = `4pt`
`bounds` = `none`
`clearance` = `4pt`
`bend-cost` = `12pt`
`pages` = `()`

Numeric nodes need no setup. Selectors for document marks/bodies/pins need setup + context. Node behavior is avoid, ignore, cut or cross; the latter two do not add visual indicators.
Guide: `docs/guides/routing-scenes.md`

deixis-route-group

NO SETUP Returns (routes, quality, stats)

Plans up to sixteen explicit sharp orthogonal page fragments together against a prepared scene.

```
deixis-route-group(((from: <a>, to: <b>),), scene: scene)
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`edges` required
`spacing` = `3pt`
`max-detour` = `25%`
`max-searches` = `64`
`scene` = `none`
`weight` = `4`
`rounds` = `2`
`max-expansions` = `50000`

Result routes retain input order. quality has before/after metrics; stats reports bounded work. No registration or painting. Complete separation/global optimality is not guaranteed. Preparing a document-derived scene still needs setup.
Guide: `docs/guides/route-groups.md`

deixis-prepare-connector

NO SETUP Returns painter request

Prepares existing geometry for native/custom paint without repeating routing. Head callbacks run here.

```
deixis-prepare-connector(path, heads: (body: "stealth"))
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`route` required
`heads` = `none`
`gap-marks` = `none`
`stroke` = `black + 1pt`
`cuts` = `none`

Place the painted request at path.origin yourself. Relative styles/custom heads may need context. It consults no document defaults, page layer or automatic crossing group.
Guide: `docs/guides/numeric-routing.md`

deixis-render-connector

NO SETUP Returns content

Paints a prepared connector request, including shafts, heads and optional decorations.

```
deixis-render-connector(request)
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`request` required

Numeric painting adds no layout dimensions; reserve the drawing area and place at the source origin. In document callbacks this is the default-painter delegation helper.
Guide: `docs/guides/numeric-routing.md`

deixis-arrowhead

NO SETUP Returns head specification

Constructs a reusable head specification; stealth is a filled triangular head with an open notch.

```
deixis-arrowhead(shape: "stealth", length: 7pt, width: 5pt)
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`shape` = `"stealth"`
`length` = `auto`
`width` = `auto`
`fill` = `auto`
`stroke` = `auto`
`trim` = `auto`
`render` = `none`

Other native shapes include triangle, vee, circle, diamond and bar. Custom render(request) returns body, trim and bounds in a tip-local +x frame. Omitted dimensions scale with the shaft. Disable endpoint heads with heads: none.

deixis-render-head

NO SETUP EXTENSION HOOK Returns content

Paints a prepared head in its local frame. A missing head or zero direction emits nothing.

```
deixis-render-head(request.heads.body, request.route.end, request.route.end-tangent)
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`head` required
`tip` required
`forward` required

Use prepared heads from the connector request, not the raw specification returned by deixis-arrowhead.

Pins And Attachments

Pin names and attachment IDs accept string/label spellings. They have independent namespaces from note IDs. A pin is a point; an attached routing object registers one live, nonbreaking frame.

Guide: `docs/guides/pins.md`

deixis-pin

SETUP REQUIRED Returns content

Publishes a geometric anchor at its declaration; padding expands bounds around that anchor.

```
deixis-pin(<anchor>, padding: 0pt)
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`name` required
`padding` = `"text"`
`scope` = `none`
`extra-data` = `(:)`

Default "text" padding follows local text metrics. Percentages have no pin-size basis. Scope namespaces pins independently of note/block identities.

deixis-attach

SETUP REQUIRED Returns content

Wraps rendered content to attach pins, scan marked tokens, or register a selectable routing node.

```
deixis-attach(id: <card>)[A card]
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`id` = `none`
`pins` = `(:)`
`inline` = `false`
`pin-padding` = `(:)`
`behavior` = `auto`
`scope` = `none`
`padding` = `auto`

id registers a routing frame with avoid behavior by default. Generated node ports are the four side midpoints. pins can define custom outlets. Explicit avoid/ignore/cut/cross also registers anonymous nodes. Pin-only attachments keep their native breakability.

Guide: `docs/guides/document-routing.md`

Scopes And Counters

Scope names and note IDs are independent. Blocks own rendering instances and counter streams; thread shares a rendering domain, while counter synchronization only shares numbering.

Guide: `docs/guides/scopes.md`

`deixis-block`
`deixis-minipage`

SETUP REQUIRED Returns content

Creates local annotation rendering/settings/counter scope around native content.

```
deixis-block(id: <section>)[Local content.]
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`id` = `auto`	`thread` = `none`
`sync-counters-with` = `auto`	`breakable` = `auto`
`width` = `auto`	`height` = `auto`
`inset` = `auto`	

Minipage maps margin to internal inset and fixes outset at 0pt. Named IDs are unique; automatic scopes stay private. Breakable blocks use per-page margin frames and collect local footnotes at the block end.

Guide: `docs/guides/blocks.md`

`deixis-note-counter`

SETUP REQUIRED Returns integer

Reads the assigned counter value for a scope and literal series. Use inside context.

```
#context deixis-note-counter()
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`target` = `0`	`series` = `"default"`

Unused streams start at zero. This is not a total annotation count. Page numbering resets do not reset note counters. Opt-in page-local display markers do not change this assigned value.

Guide: `docs/guides/counters.md`

`deixis-update-note-counter`

SETUP REQUIRED Returns content

Sets a stream to an integer or applies current ⟶ integer. The next original increments that value.

```
deixis-update-note-counter(0)
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`value` required	`target` = `0`
`series` = `"default"`	

Bodies, references and reuse never increment. Earlier notes keep assigned numbers. Endnote wrappers use the "endnote" series; select it explicitly for their resets.

Guide: `docs/guides/counters.md`

References And Custom Views

Queries return safe semantic records and summaries instead of replaying live annotation content. An annotation remains one entry even when several marks share its body.

Guide: `docs/guides/queries.md`

deixis-ref

SETUP REQUIRED Returns content

References one or more annotations, retaining original marker styling and native fallback for ordinary labels.

```
deixis-ref(<detail>)
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`..targets` variadic
`form` = `"normal"`
`supplement` = `auto`
`marker-style` = `auto`

@detail uses the native bridge installed by setup. Repeated reference targets deduplicate; compatible numeric ranges can collapse. Explicit marker-style overrides formatting.

Guide: `docs/guides/references.md`

deixis-query-notes

SETUP REQUIRED Returns array of records

One safe structured record per selected annotation. Page queries follow native body order; the unfiltered catalogue follows source order. Call inside context.

```
#context deixis-query-notes(page: here().page())
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`target` = `auto`
`include-types` = `auto`
`include-unpaired` = `auto`
`before` = `none`
`order` = `auto`
`thread` = `auto`
`series` = `auto`
`after` = `none`
`page` = `auto`

Page queries include standalone bodies and require complete unique live spans. order: "source" / "body" and include-unpaired override automatic defaults. Each record retains all shared marks, safe summaries and optional body metadata; no live body content is returned.

deixis-note-outline

SETUP REQUIRED Returns content

Ready-made linked list over the annotation catalogue; custom callbacks can replace summaries or presentation.

```
deixis-note-outline()
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`title` = `[List of Notes]`
`include-unpaired` = `false`
`summary` = `auto`
`include-types` = `auto`
`group-by` = `none`

One entry per annotation, with links to all marked pages. Use query-notes for cards/tables or occurrence indexes. data can supply a custom title/status.

Guide: `docs/guides/outlines.md`

deixis-plain

SETUP REQUIRED Returns content

Shows repeated content with its marked passages but without creating marks, bodies or connections again.

```
deixis-plain(heading.body)
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`body` required

Native outlines apply it to heading and caption entries automatically. Use it for running headers and custom tables of contents that repeat annotated headings.

Guide: `docs/guides/outlines.md`

Rendering Extension Points

These hooks consume supplied records or content; they do not create a registered annotation. When used in live document annotations, keep setup active and preserve destination/content helpers. Relative typography and backlinks may require context.

Guide: `docs/guides/custom-renderers.md`

`deixis-native-render-single`
`deixis-aligned-render-single`

CONDITIONAL **EXTENSION HOOK** Returns content

Built-in single-body layouts. Native keeps a natural text flow; aligned accepts a shared marker column.

```
deixis-native-render-single(note, ..args.named())
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`data` required `inner-width` = `auto`

Use supplied data unchanged when delegating. Aligned additionally accepts max-marker-width/max-marker-gap. The container and body-style come from the body record.

`deixis-native-render-group`
`deixis-aligned-render-group`
`deixis-grid-render-group`

CONDITIONAL **EXTENSION HOOK** Returns content

Stack bodies, align stacked markers, or lay markers/content directly in two grid columns.

```
deixis-native-render-group(notes, ..args.named())
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`notes-data` required `gap` = `auto`

Native/aligned groups delegate to each single renderer. Grid bypasses single renderers and containers. Empty groups produce no content. Supplied records may contain live document behavior.

Guide: `docs/guides/collected-notes.md`

`deixis-generate-body-meta`
`deixis-generate-body-marker`
`deixis-generate-body-content`
`deixis-generate-backlinks`

CONDITIONAL **EXTENSION HOOK** Returns content

Composable helpers for a custom body renderer: one destination, a marker, live content span, and return links.

```
[#deixis-generate-body-marker(note)#deixis-generate-body-content(note)]
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`data` required

Emit meta OR marker once; marker already includes meta. Content publishes start/end spans and preserves nested palette ownership. Measurement records suppress live metadata. Backlinks read document origins and need setup/context.

`deixis-plain-container`
`deixis-rect-container`
`deixis-alert-container`

NO SETUP EXTENSION HOOK Returns content

Body wrappers: identity, native rect alias, or a left-bar alert. Container arguments are forwarded separately from shared styles.

```
deixis-inset-note(container-func: "tinted", container-args: (inset: 6pt))[Body]
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`body` required `..args` variadic

container-func accepts names plain, rect, alert, stripe and tinted, or a custom function. The first three select these existing exports; stripe and tinted are named-only accent presets. Explicit stroke/fill/radius and container-args override preset defaults. rect-container has native rect parameters. Alert doubles scalar border thickness on the left; native edge maps remain literal.

Guide: `docs/guides/containers.md`

`deixis-rect-region`
`deixis-ellipse-region`

NO SETUP EXTENSION HOOK Returns content

Region-shape callbacks for rectangular or elliptical bounds.

```
deixis-rect-region(width: 40pt, height: 20pt, fill: teal)
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`..args` variadic

Shape records provide width, height, fill/stroke and rectangle radius. Region registration/fragment selection belongs to the calling mark, not these painters. Use region marks with stroke: none for a highlight; add region-shape: deixis-ellipse-region for an ellipse.

Guide: `docs/guides/regions.md`

Low-Level Layout Hooks

Advanced hooks use supplied measured records and geometry. Most documents should choose note adapters, container/render callbacks, or routing APIs instead. Native layout and document lookup are separate responsibilities.

Guide: `docs/guides/rendering.md`

`deixis-margin-note-exact-layout`
`deixis-margin-note-adaptive-layout`
`deixis-margin-note-flow-layout`

CONDITIONAL **EXTENSION HOOK** Returns content

Numeric margin adapter hooks: exact ignores collisions; adaptive packs; flow distributes available space.

SELECTED PARAMETERS · SOURCE DEFAULTS

`notes` required
`bottom-bound` required
`text-width` required
`left-space` required
`current-page` required
`top-bound` required
`start-x` required
`end-y` required
`right-space` required

All share min-margin-width, pushed-notes and extra arguments. Supply accepted page geometry and prepared notes. They paint locally through the layout adapter. These hooks are not the body layer: "flow" switch.

`deixis-place-anchored`

SETUP REQUIRED **EXTENSION HOOK** Returns content

Immediate placement relative to selected pin bounds on the call's page.

SELECTED PARAMETERS · SOURCE DEFAULTS

`body` required
`dy` = `0pt`
`pin` = `none`
`dx` = `0pt`
`anchor` = `bottom + right`
`body-size` = `auto`

Pin lookup reads document geometry. Native placement can disturb flow.

Guide: `docs/guides/pins.md`

`deixis-absolute-place`

CONDITIONAL **EXTENSION HOOK** Returns content

Immediate placement in physical page coordinates, ignoring margins.

```
deixis-absolute-place(top + left, dx: 30pt, dy: 40pt)[Content]
```

SELECTED PARAMETERS · SOURCE DEFAULTS

`..args` variadic

Accepts content, or alignment + content; dx/dy default to 0pt. Fixed page dimensions need no setup. An auto page axis needs setup-provided page frames for accepted dimensions. This is placement, not dimension reservation.

Integration Checklist

- Configure native page layers before setup; reapply setup after replacing them. Host-owned overlays still need initialization. Changing dimensions/margins alone does not need another setup call.
- Use flow for long footnotes. Overlays cannot create pages. Immediate margin placement has no collision packing and native placement may disturb paragraphs.
- A body declaration controls formatting and adapter-specific allocation, even with forward identity lookup. Collected bodies must be printed to be endpoints.
- Multi-page region highlights need overlays. Arbitrary native columns/fragments are not automatically discovered. Deep deferred contexts have a finite native layout-pass budget; do not replace missing geometry with guessed bounds.
- In custom renderers, emit destination/marker once and live body content once. Query summaries are safe; replaying raw executable body content is not.
- For Touying, use the frame identity/reveal recipes. For migration from 0.1.1, read `docs/guides/compatibility.md`.

Read the task-oriented index in `docs/README.md`. Source examples live in `examples/` and `docs/examples/`; the worked pairs name their editable sources. This checkout is unreleased: install its 0.2.0 preview override to compile the examples. Build this source with `python3 scripts/render-manual.py`.
Bug reports and source: https://github.com/inspiros/typst-deixis.